

Power Consumption Forecasting Project

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Project Goals

- Predict the power consumption of Tetouan City using weather, flow, and time-based variables, then compare several statistical and machine learning models to see which works best.
 - We will build separate models for Zone 1, Zone 2, and Zone 3, because each zone has its own consumption target.

Main Questions

1. How accurately can we predict power consumption for each zone using the available features?
2. Which variables matter most for prediction?
3. Do nonlinear models outperform linear regression?
4. Does adding time features like hour, day of week, or month improve performance?



Data To Use

Dataset: Tetuan City power consumption.csv

Predictors: Datetime, Temperature, Humidity, Wind Speed, general diffuse flows, diffuse flows

Targets: Zone 1 Power Consumption, Zone 2 Power Consumption, Zone 3 Power Consumption

Workflow

Step 1: Load &
Inspect Data

Step 2: Exploratory
Data Analysis

Step 3: Feature
Engineering

Step 6: Fit
Improved Models

Step 5: Fit Baseline
Models

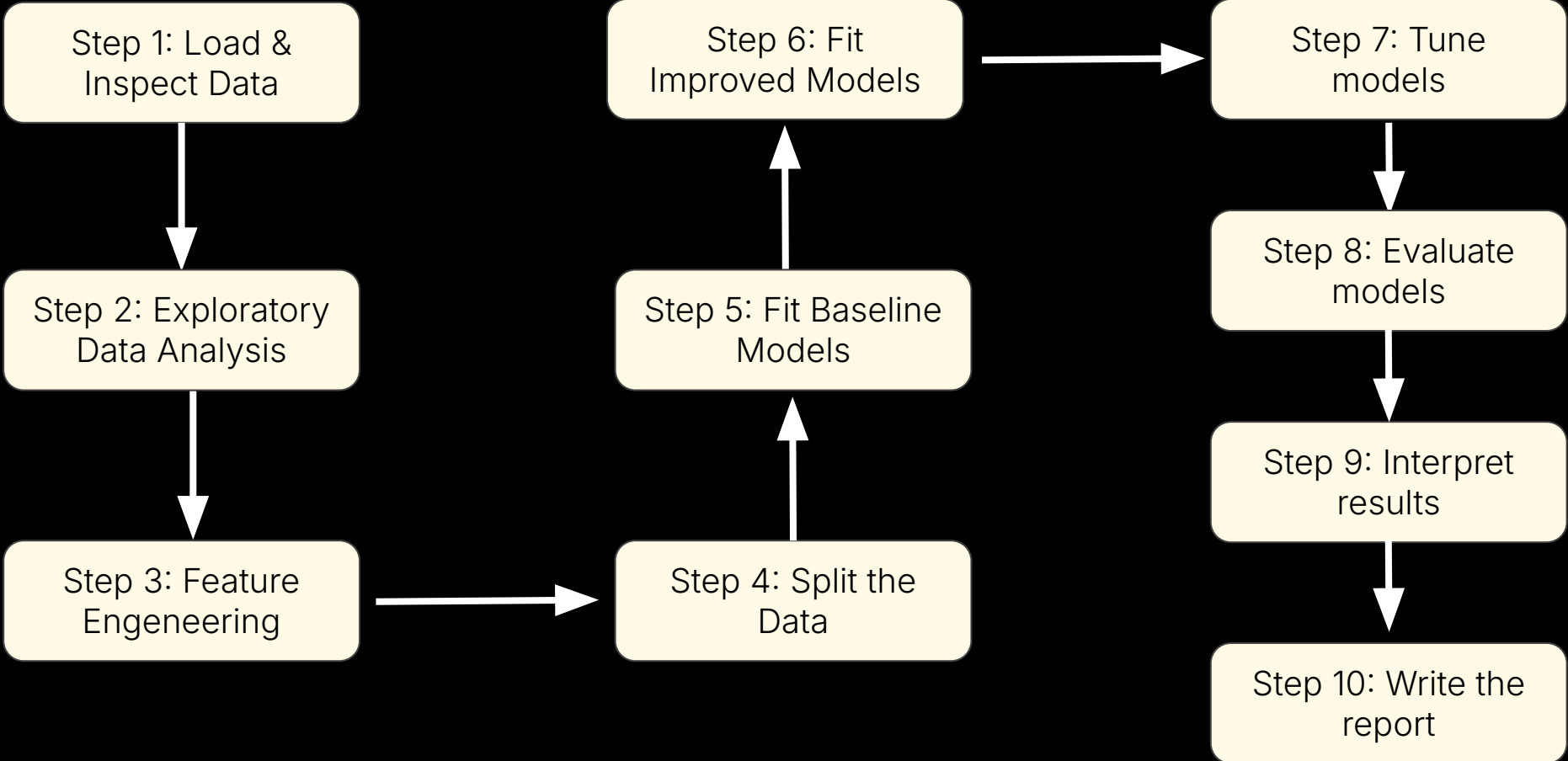
Step 4: Split the
Data

Step 7: Tune
models

Step 8: Evaluate
models

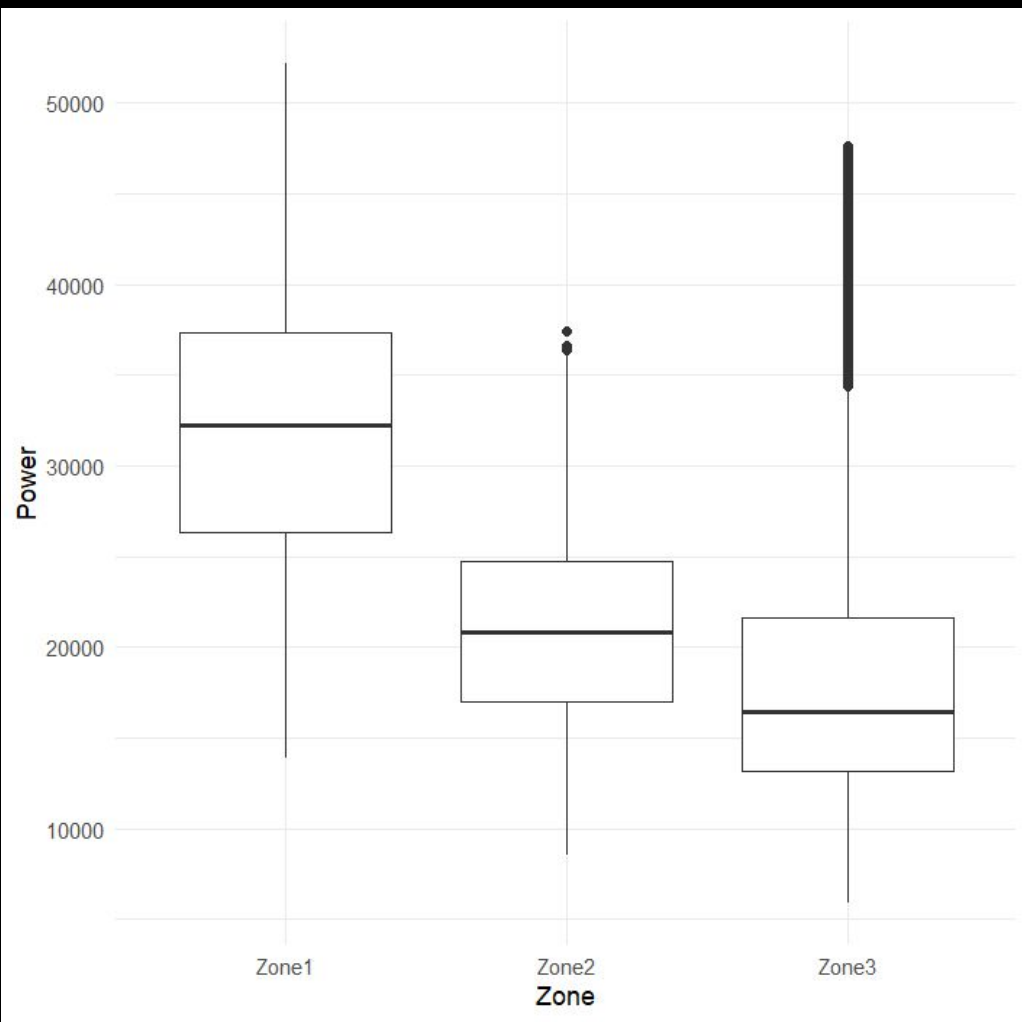
Step 9: Interpret
results

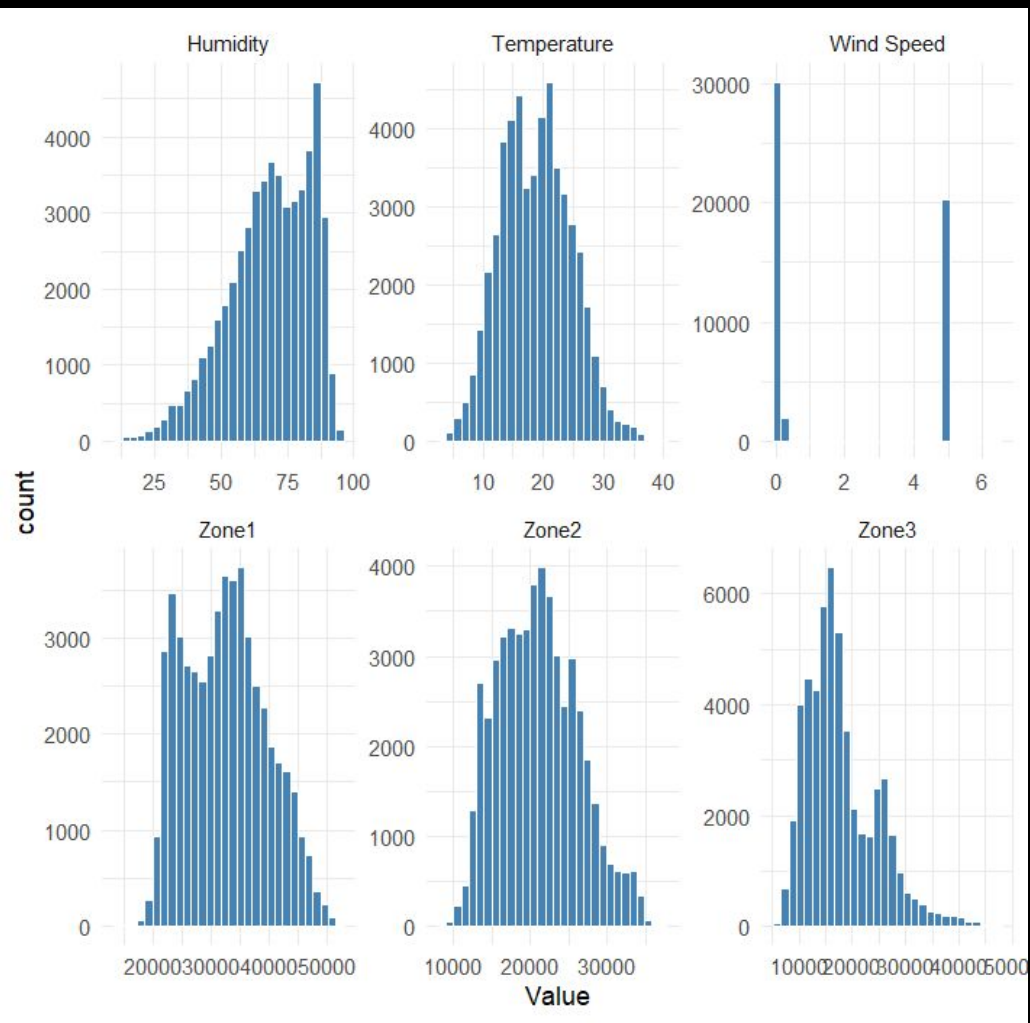
Step 10: Write the
report

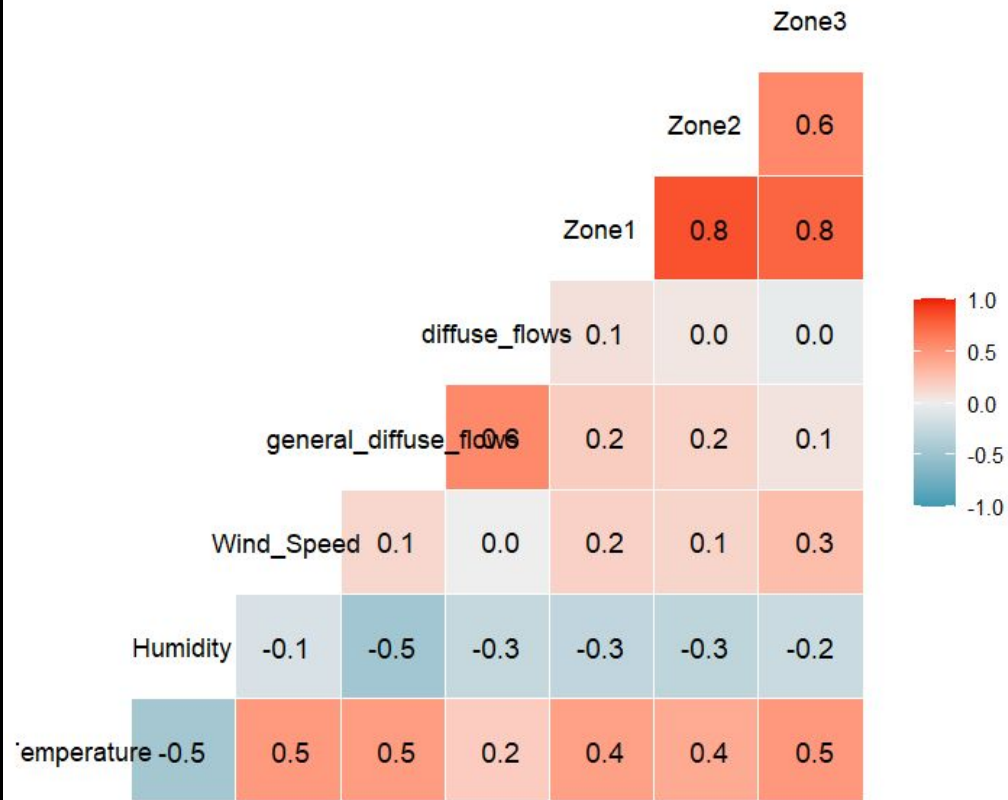


Power Consumption Over Time









Model Comparison

Model	Zone	RMSE	MAE	R2
<chr>	<chr>	<dbl>	<dbl>	<dbl>
1 Ridge	Zone 1	4199.	3863.	0.536
2 Lasso	Zone 1	4690.	4142.	0.422
3 Elastic Net	Zone 1	4640.	4104.	0.434
4 Random Forest	Zone 1	4647.	4300.	0.432
5 XGBoost	Zone 1	4814.	4164.	0.390
6 Ridge	Zone 2	2208.	1894.	0.838
7 Lasso	Zone 2	180.	146.	0.999
8 Elastic Net	Zone 2	312.	251.	0.997
9 Random Forest	Zone 2	5159.	4619.	0.115
10 XGBoost	Zone 2	139.	57.7	0.999
11 Ridge	Zone 3	3002.	2701.	0.172
12 Lasso	Zone 3	230.	210.	0.995
13 Elastic Net	Zone 3	392.	378.	0.986
14 Random Forest	Zone 3	2484.	2159.	0.433
15 XGBoost	Zone 3	359.	145.	0.988

Best Models

Model	Zone	RMSE	MAE	R2
<chr>	<chr>	<dbl>	<dbl>	<dbl>
1 Ridge	Zone 1	4199.	3863.	0.536
2 XGBoost	Zone 2	139.	57.7	0.999
3 Lasso	Zone 3	230.	210.	0.995



Thank You!

Questions?